Fundamentals Chapter 11

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1a. First, we need to show that $|x-y| = 0$ iff $x = y$. If $x = y$ then $x-y = 0$, so $|x - y| = 0$. To show that if $|x - y| = 0$ then $x = y$, assume $x \neq y$. Then $y = x + z$ for some nonzero z, so $|x - y| = |x - (x + z)| = | - z| \neq 0$, which is a contradiction. So $x = y$.

Now, we must show that $|x-y| \leq |x-z|+|y-z|$. To do this, we can define $a = x - z$ and $b = z - y$ so that the inequality becomes

$$|a + b| \leq |a| + |b|.$$

Now, we square both sides and simplify, leaving us with

$$a^2 + b^2 + 2ab \leq a^2 + b^2 + 2|ab|,$$

$$ab \leq |ab|,$$

which we know is true. So the original inequality is true as well and $d(x, y) = |x - y|$ is a metric.

1b. We want to show that $d(x, y) = \|x - y\|$ is a metric on $\mathbb{R}^2$. From a similar argument to before, $\|x - y\| = 0$ iff $x = y$.

Now, to prove the triangle inequality, we let $a = \|x - z\|$ and $b = \|z - y\|$, so that the desired result is

$$\|a + b\| \leq \|a\| + \|b\|.$$

Let $a = (a_1, a_2)$ and $b = (b_1, b_2)$. Because both sides of the inequality are always nonnegative, we can square both sides to obtain

$$\|a + b\|^2 \leq \|a\|^2 + \|b\|^2,$$

or

$$(a_1 + b_1)^2 + (a_2 + b_2)^2 \leq (a_1^2 + a_2^2) + (b_1^2 + b_2^2) + 2\sqrt{(a_1^2 + a_2^2)(b_1^2 + b_2^2)}.$$

Expanding and simplifying gives us

$$a_1b_1 + a_2b_2 \leq \sqrt{(a_1^2 + a_2^2)(b_1^2 + b_2^2)}.$$

Assume the LHS is negative. Then, since the square root is positive, the inequality holds. So assume the LHS is nonnegative. This lets us square both sides:

$$(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2).$$

Simplifying and then factoring gives us

$$(a_1b_2 - a_2b_1)^2 \geq 0,$$

which we know is always true. Therefore, the original inequality is also true, and $d(x, y) = \|x - y\|$ is a metric on $\mathbb{R}^2$.

2. To prove that $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$, we can consider six different cases corresponding to the six different orderings of x, y, z:

Case 1: $z \leq y \leq x$. In this case, the inequality is equivalent to $x - y \leq x - z + 1$, or $z - y \leq 1$, which is true because $z \leq y$.

Case 2: $y \leq z \leq x$. The inequality is equivalent to $x - y \leq x - z + z - y$, or $0 \leq 0$ which is true.

Case 3: $z \leq x \leq y$. The inequality is $1 \leq x - z + 1$, or $z \leq x$ which is true.

Case 4: $x \leq z \leq y$. The inequality is $1 \leq 1 + 1$, which is true.

Case 5: $y \leq x \leq z$. The inequality is $x - y \leq 1 + z - y$, or $x \leq 1 + z$, which is true.

Case 6: $x \leq y \leq z$. The inequality is $1 \leq 1 + z - y$, or $y \leq z$, which is true.

We can see that $\rho(x, y) \neq \rho(y, x)$ because one of them is $|x-y|$ and the other is 1. Since $\rho(x, y) = 0$ iff $x = y$, this can only be true if $\rho(x, y) \leq \rho(x, z)+\rho(y, z)$ is false.

3a. By the definition of d, $d(x, y) = 0$ iff $x = y$. To show the triangle inequality, first consider the case where $x = y$. Here, the inequality simplifies to $0 \leq 2d(x, z)$, which is true.

Now, assume $x \neq y$. Then the inequality simplifies to $1 \leq d(x, z) + d(y, z)$. This is only false if $d(x, z)$ and $d(y, z)$ are both 0, which would imply that $x = z$ and $y = z$ so $x = y$, which is a contradiction. So the inequality is true.

3b. In the case where $x = y$, $d(x, y) = 0$ so it is $\leq d(x, z)$ and $d(z, y)$. If $x \neq y$, then $d(x, z)$ and $d(y, z)$ can't both be 0 at the same time, so $d(x, y) \leq \max(d(x, z), d(y, z)) = 1$.

5a. Let $N > \frac{2}{\varepsilon}$. Then, if $m, n \geq N$,

$$\left|\frac{1}{n} - \frac{1}{m}\right| \leq \left|\frac{1}{n}\right| + \left|\frac{1}{m}\right| = \frac{1}{n} + \frac{1}{m} \leq \frac{2}{N} < \varepsilon.$$

5b. Let $N > \frac{2}{\varepsilon}$. Then, if $m, n \geq N$,

$$\left|\frac{\sin(n)}{n} - \frac{\sin(m)}{m}\right| \leq \left|\frac{\sin(n)}{n}\right| + \left|\frac{\sin(m)}{m}\right| \leq \frac{1}{n} + \frac{1}{m} \leq \frac{2}{N} < \varepsilon.$$

5c. We can rewrite $\frac{2n^2+1}{n^2}$ and $\frac{2n+1}{n}$ as $2 + \frac{1}{n^2}$ and $2 + \frac{1}{n}$ respectively. Let $m, n \geq N > \max\left(\frac{2}{\varepsilon}, \sqrt{\frac{2}{\varepsilon}}\right)$.

In the case where m, n are odd:

$$\left|2 + \frac{1}{n^2} - 2 - \frac{1}{m^2}\right| \leq \frac{1}{n^2} + \frac{1}{m^2} < \frac{2}{N^2} < \epsilon.$$

In the case where m is odd and n is even:

$$\left|2 + \frac{1}{n} - 2 - \frac{1}{m^2}\right| \leq \frac{1}{n} + \frac{1}{m^2} < \frac{N+1}{N^2} \leq \frac{2}{N} < \epsilon.$$

The case where m is even and n is odd is essentially the same as the above case. In the case where m, n are even:

$$\left|2 + \frac{1}{n} - 2 - \frac{1}{m}\right| \leq \frac{1}{n} + \frac{1}{m} < \frac{2}{N} < \epsilon.$$

5d. The sequence is bounded above by π and monotone increasing. Therefore, it is Cauchy.

6a. We know that, given any $\varepsilon > 0$, there exists N such that, if $n, m > N$, then $|x_n - x_m| < \varepsilon$. We wish to show that

$$||x_n| - |x_m|| < \varepsilon$$

as well. We claim that $||x_n| - |x_m|| \leq |x_n - x_m|$. To show this, we can square both sides and rearrange:

$$(|x_n| - |x_m|)^2 \leq (x_n - x_m)^2,$$

$$x_n^2 + x_m^2 - 2|x_n x_m| \leq x_n^2 + x_m^2 - 2x_n x_m,$$

$$x_n x_m \leq |x_n x_m|,$$

which is true. So our claim is true, and if $n, m > N$, then

$$||x_n| - |x_m|| \leq |x_n - x_m| < \varepsilon.$$

6b. Fix $\varepsilon > 0$. We know that there exists N_1 such that if $n, m > N_1$ then $|x_n - x_m| < \varepsilon/2$, and there exists N_2 such that if $n, m > N_2$ then $|y_n - y_m| < \varepsilon/2$. Let $N = \max(N_1, N_2)$. If $n, m > N$, then

$$|(x_n + y_n) - (x_m + y_m)| \leq |x_n - x_m| + |y_n - y_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

6c. Since x_n and y_n are Cauchy, they are bounded, so $\sup |x_n|$ and $\sup |y_n|$ exist. For all $\varepsilon > 0$, there exist N_1 and N_2 such that

$$|x_n - x_m| < \frac{\varepsilon}{2 \sup |y_n|} \text{ if } n, m > N_1,$$

$$|y_n - y_m| < \frac{\varepsilon}{2 \sup |x_n|} \text{ if } n, m > N_2.$$

If $n, m > \max(N_1, N_2)$ then

$$|x_n y_n - x_m y_m| = |(x_n y_n - x_n y_m) + x_n y_m - x_m y_m|$$

$$= |x_n(y_n - y_m) + y_m(x_n - x_m)| \leq |x_n(y_n - y_m)| + |y_m(x_n - x_m)|$$

$$= |x_n||y_n - y_m| + |y_m||x_n - x_m| < |x_n| \frac{\varepsilon}{2 \sup |x_n|} + |y_m| \frac{\varepsilon}{2 \sup |y_n|}$$

$$\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $x_n y_n$ is Cauchy.

7a. We need to show that there exists $\varepsilon > 0$ such that for all N, there exist $m, n > N$ such that $|n - m| \geq \varepsilon$. Let $\varepsilon = \frac{1}{2}$. Given $n, m > N$ where $m \geq n + 1$, $|n - m| \geq 1 > \varepsilon$.

7b. Fix ε, and let $m = N + 1$ and $n \geq (\varepsilon + \sqrt{N+1})^2$. Then

$$\left|\sqrt{n} - \sqrt{m}\right| = \sqrt{n} - \sqrt{N+1} \geq \varepsilon + \sqrt{N+1} - \sqrt{N+1} = \varepsilon.$$

7c. Let $\varepsilon = \frac{1}{2}$. Given any N, choose odd n and even m. Then $|c_n - c_m| = 1 \geq \varepsilon$.

8. Let $a = x - z$ and $b = z - y$. Then we wish to prove that

$$\|a + b\| \leq \|a\| + \|b\|.$$

Squaring both sides, we get

$$\|a + b\|^2 \leq \|a\|^2 + \|b\|^2 + 2\|a\|\|b\|,$$

$$\langle a + b, a + b \rangle \leq \langle a, a \rangle + \langle b, b \rangle + 2\|a\|\|b\|.$$

We can further simplify the LHS as:

$$\langle a + b, a + b \rangle = \langle a, a + b \rangle + \langle b, a + b \rangle$$

$$= \overline{\langle a+b, a\rangle} + \overline{\langle a+b, b\rangle} = \overline{\langle a, a\rangle + \langle b, a\rangle} + \overline{\langle a, b\rangle + \langle b, b\rangle}$$
$$= \langle a, a\rangle + \langle a, b\rangle + \langle b, a\rangle + \langle b, b\rangle.$$

Plugging this into the inequality and canceling terms, we are left with

$$\langle a, b\rangle + \langle b, a\rangle \leq 2\|a\|\|b\|.$$

By Cauchy-Schwarz, $\langle a, b\rangle \leq \|a\|\|b\|$ and $\langle b, a\rangle \leq \|a\|\|b\|$. Adding those two yields the desired inequality.

10. Let's say $d(a, b) = d_1$, $d(a, c) = d_2$, and $d(b, c) = d_3$. Define $f : \{a, b, c\} \to \mathbb{R}^2$ so that $f(a) = (0, 0)$ and $f(b) = (d_1, 0)$. Now, $f(c)$ needs to be chosen such that $\|f(c) - f(a)\| = d_2$ and $\|f(b) - f(a)\| = d_3$. So $f(c)$ is the intersection between the circle centered at $f(a)$ with radius d_2 and the circle centered at $f(b)$ with radius d_3. Since $d_1 \leq d_2 + d_3$, this point exists, and we have found an isometry.

11a. To show that all elements of D_n are isometries, we can start by proving that reflections across any line preserve distance. Let $a, b \in \mathbb{R}^2$ and denote a reflection about line ℓ by f_ℓ. We can denote a, b by points A, B, and $f_\ell(a), f_\ell(b)$ by A', B'. Also, define X to be the intersection of ℓ and AA', and Y to be the intersection of ℓ and BB'.

Our goal is to show that $AB = A'B'$. In the case where A, B are on the same side of ℓ, by the Pythagorean theorem, we have that

$$AB^2 = (AX - BY)^2 + XY^2,$$
$$A'B'^2 = (A'X - B'Y)^2 + XY^2.$$

Since A', B' are reflections about ℓ, $AX = A'X$ and $BX = B'X$, so $AB^2 = A'B'^2$, or $AB = A'B'$.

In the case where A, B are on opposite sides of ℓ, we have that

$$AB^2 = (AX + BY)^2 + XY^2,$$
$$A'B'^2 = (A'X + B'Y)^2 + XY^2,$$
$$AB^2 = A'B'^2.$$

So reflections preserve distance.

Now, we show that rotations preserve distance. Let $a, b \in \mathbb{R}^2$, and f_θ denote rotation about some point O by θ radians. Denote a, b by points A, B and $f_\theta(a), f_\theta(b)$ by A', B'. Consider the triangles OAB and $OA'B'$. Since f_θ is a rotation, $OA = OA'$ and $OB = OB'$. Also, since $\angle AOB = \angle A'OB' = \theta$, the two triangles are congruent, so $AB = A'B'$. Therefore, rotations preserve distance, and all elements of D_n are isometries.

11b. Other isometries of the plane are translations or combinations of reflections, rotations, and translations.